

Marton Mester

Mars Climate Modeling Center
NASA Ames Research Center
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EDUCATION	University College London , London, UK Ph.D. in Applied Mathematics 2016-2021 Advisors: J. Gavin Esler (principal), Robb McDonald (secondary)		
	University of Cambridge, Trinity Hall , Cambridge, UK B.A. in Mathematics 2012-2015 (Class I,I,II/1) MMath 2015-2016 (Honours)		
LANGUAGES	English, Hungarian		
NATIONALITY	Hungarian		
TECHNICAL SKILLS	MATLAB, Fortran		
RESEARCH EXPERIENCE	NASA Ames Research Center , Moffett Field, CA, USA NPP Research Fellow 2024 - 2026 Project: Influence of Mars' Topography on Traveling Weather Systems and Cross-Equatorial Dust Storms		
	Complutense University of Madrid , Madrid, Spain Postdoctoral Research Fellow 2021 - 2023 Project: Dynamical forcing and triggering mechanisms of sudden stratospheric warmings		

University College London, London, UK
Research Assistant (Ph.D.), Program in Applied Mathematics 2016 - 2021
Title: Stochastic and Dynamical Models of Sudden Stratospheric Warmings

University of Cambridge, Cambridge, UK
Cambridge Summer Research in Mathematics (CSRIM) Programme Summer 2015
Title: Optimal Vortex States for Planetary-scale Rossby-wave Amplification
Advisor: Peter Hitchcock

TEACHING EXPERIENCE **University College London**, London, UK
Undergraduate courses: Fluid mechanics: Demonstrator, tutorial and problem class teacher, examiner

PROFESSIONAL ACTIVITIES **University College London**, London, UK
UCL Postgraduate Fluid Dynamics Seminar series, organiser 2017-2019

NASA
NSPIRES panelist 2025

PEER REVIEW ACTIVITY Nature communications: earth & environment, Journal of Climate, Journal of the Atmospheric Sciences, Icarus, JGR: Planets, Geophysical Research Letters, Chaos, Remote Sensing

CONFERENCE ABSTRACTS AGU25, New Orleans, LA, USA 19/12/25
Radiatively Coupled Equatorial Waves in Mars' Aphelion Cloud Belt: Wavenumber-Frequency Analysis in Mars GCM Simulations

Bay Area Planetary Science Conference 2025, Stanford University, Stanford, CA, USA 24/09/25
The Influence of Mars' Continental-Scale Topographic Structures on its Hemispheric Asymmetry in Baroclinic Wave Activity

The Tenth International Conference on Mars, Pasadena, CA, USA 22-25/07/24
- *Influence of the Hellas and Argyre impact basins on stationary and transient waves in Mars' Southern Hemisphere*
- *Interpreting surface pressure observations with a global climate model*
- *Large-scale extratropical weather systems in the southern hemisphere of Mars*

24th Conference on AOFD, Burlington, VT, USA 24-28/06/24
- *On the Triggering Mechanisms of Vortex-Splitting Stratospheric Sudden Warmings in Idealized Model Simulations*
- *The Influence of the Hellas and Argyre Impact Basins on Mars' Stationary and Travelling Waves*

7th SPARC General Assembly, Reading, UK 24-28/10/22
Signatures of resonantly triggered vortex splitting sudden warmings in idealized model

experiments

Workshop Mars V, Les Houches, France 26/04/22
Control of the Oscillations of the Martian Northern Hemisphere Polar Vortex by the Hadley Cell and Topographic Forcing

5th Chianti Topics - International Focus Workshop, Bagno a Ripoli, Firenze, Italy 20/04/22
Oscillations of the Martian Northern Hemisphere Polar Vortex, a Contour Dynamics Perspective

22nd Conference on AOFD, Portland, ME, USA 24/06/19
Dynamical Elliptical Diagnostics of the Antarctic Polar Vortex

Dynamics of Rotating Fluids meeting, University of Oxford, Oxford 21/09/18
Noise-induced split-type stratospheric sudden warmings

Summer Undergraduate Research Presentations, University of Cambridge, Cambridge 12/10/15
Optimal vortex states for planetary-scale Rossby wave amplification

PUBLICATIONS

10. M. Mester, J. G. Esler, P. H. Hitchcock: **Control of extreme stratospheric vortex events in idealized GCM simulations**
in preparation
9. M. Mester, A. de la Camara, P. Zurita-Gotor: **On the Triggering Mechanisms of Vortex-Splitting Stratospheric Sudden Warmings in Idealized Model Simulations**
in preparation
8. R. J. Wilson, M. Mester: **Isolating Diurnal and Semidiurnal Migrating Tides from Martian Surface Pressure Observations**
in preparation
7. M.A. Kahre, R. A. Urata, R. J. Wilson, C. M. Batterson, A. Brecht, K. Steakley, A. Kling, T. Bertrand, C. E. Harman, L. Gkouvelis, M. Mester: **Documentation of the NASA Ames FV3-Based Mars Global Climate Model: Simulations of the Current Climate and Water Cycle**
under review
6. M. Mester, R. J. Wilson, M. A. Kahre, 2026: **Radiatively coupled equatorial waves in Mars' Aphelion Cloud Belt: Wavenumber-frequency analysis in Mars GCM Simulations**
Icarus, Volume 449, 2026, 116967, <https://doi.org/10.1016/j.icarus.2026.116967>

5. M. Mester, R. J. Wilson, J. L. Hollingsworth, M. A. Kahre, 2025: **The influence of Mars' continental-scale topographic structures on its hemispheric asymmetry in baroclinic wave activity**
Icarus, Volume 444, 2026, 116833, <https://doi.org/10.1016/j.icarus.2025.116833>
4. M. Mester, R. J. Wilson, M. A. Kahre, 2025: **Transient and tidal wave precursors and atmospheric impacts of Mars' flushing storm initiated regional storms in reanalysis data**
Icarus, Volume 441, 2025, 116690, <https://doi.org/10.1016/j.icarus.2025.116690>
3. M. Mester, 2022: **Control of the oscillations of the martian Northern Hemisphere polar vortex by the Hadley cell and topographic forcing**
Icarus, Volume 382, 2022, 115019, <https://doi.org/10.1016/j.icarus.2022.115019>
2. M. Mester, J. G. Esler, 2020: **Dynamical Elliptical Diagnostics of the Antarctic Polar Vortex**
J. Atmos. Sci., 77, 1167-1180, <https://doi.org/10.1175/JAS-D-19-0232.1>
1. J. G. Esler, M. Mester, 2019: **Noise-induced vortex-splitting stratospheric sudden warmings**
Quart. J. Roy. Meteor. Soc. 2019; 145: 476-494. <https://doi.org/10.1002/qj.3443>